

What's so special about Euler's number e ?

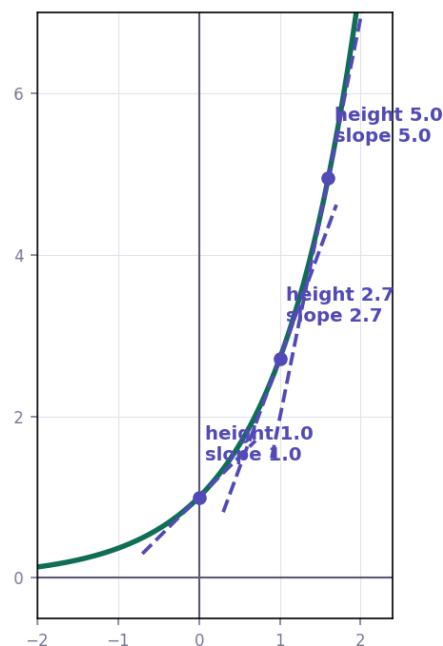
The function that is its own derivative.

THE IDEA

Exponential functions have a special property: their rate of growth is proportional to their current size. Among all of them, exactly one has the growth rate **equal** to its size — the one with base **$e \approx 2.71828$** .

The defining fact: $d/dx (e^x) = e^x$ — at every point, the height of the curve equals its slope.

$y = e^x$ — the curve that is its own derivative



THE FORMULAS

Function	Derivative
e^x	e^x
$\ln x$	$1/x$
a^x	$a^x \cdot \ln(a)$
$e^{g(x)}$	$e^{g(x)} \cdot g'(x)$ (chain rule)

WORKED EXAMPLES

CHAPTER 5 — WORKED EXAMPLES

exponentials and the number e

THE KEY FACTS

$$\frac{d}{dx} (e^x) = e^x$$

the only function that is its own derivative

$$\frac{d}{dx} (\ln x) = 1/x$$

$$\frac{d}{dx} (a^x) = a^x \cdot \ln(a)$$

EXAMPLE 1 $y = e^{3x}$

chain rule : inside $g = 3x \rightarrow g' = 3$

$$dy/dx = e^{3x} \cdot 3 = 3e^{3x}$$

EXAMPLE 2 $y = e^{(x^2)}$

inside $g = x^2 \rightarrow g' = 2x$

$$dy/dx = e^{(x^2)} \cdot 2x = 2x \cdot e^{(x^2)}$$

EXAMPLE 3 $y = 2^x$

$$dy/dx = 2^x \cdot \ln(2)$$

ML link : e^x is inside the **sigmoid** and **softmax** functions,
and $\ln x$ is inside **cross-entropy loss**.

Why it matters for ML: e is everywhere in machine learning — inside the **sigmoid** function $1/(1+e^{-x})$, inside **softmax**, and its inverse $\ln$ appears in **cross-entropy loss**. Being able to differentiate these is how those losses get trained.

FORMULAS FROM THIS CHAPTER

$d/dx (e^x)$	e^x
$d/dx (\ln x)$	$1/x$
$d/dx (a^x)$	$a^x \cdot \ln a$
With chain rule	$d/dx e^{g(x)} = e^{g(x)} \cdot g'(x)$

TEST — Chapter 5

1. Differentiate $y = e^{5x}$. $\rightarrow 5e^{5x}$
2. Differentiate $y = e^{(3x^2)}$. $\rightarrow e^{(3x^2)} \cdot 6x$
3. Differentiate $y = \ln(x) + 3^x$. $\rightarrow 1/x + 3^x \cdot \ln(3)$